Qn	Ans	Solution
18	D	At lowest frequency, $L = \frac{1}{4}\lambda_1$ $\lambda_1 = 4L$ $f_1 = \frac{v}{\lambda_1} = \frac{v}{4L}$ At next highest frequency, $L = \frac{3}{4}\lambda_2$ $\lambda_2 = \frac{4}{3}L$ $f_2 = \frac{v}{\lambda_2} = \frac{3v}{4L} = 3\frac{v}{4L} = 3f_1$ At third highest frequency, $L = \frac{5}{4}\lambda_3$ $\lambda_3 = \frac{4}{5}L$ $f_3 = \frac{v}{\lambda_3} = \frac{5v}{4L} = 5\frac{v}{4L} = 5f_1$ So the frequencies that can be produced by the instrument are $f_1,\ 3f_1,\ 5f_1,\ 7f_1,\ ...$
19	A	Option A: Correct